

作业9 4-1 (1)(3)(5)
4-2 (1)(3)(5)
4-3
4-4 (1)(2)
4-5

4-1

$$\begin{aligned} a_k &= \frac{1}{6} \sum_{n=0}^5 x[n] e^{-j\frac{2\pi}{6}kn} \\ &= \frac{1}{6} \sum_{n=0}^5 e^{-j\frac{\pi}{3}kn} \\ &= \frac{1}{6} (1 + e^{-j\frac{\pi}{3}k} + e^{-j\frac{2\pi}{3}k} + e^{-j\pi k}) \\ &= \frac{1}{6} [1 + e^{-j\frac{\pi}{3}k} + e^{-j\frac{2\pi}{3}k} + e^{-j\pi k}] \end{aligned}$$

(3)

$$\begin{aligned} a_k &= \frac{1}{4} (2e^{-j\frac{\pi}{4}k} + 4e^{-j\frac{\pi}{2}k}) \\ &= \frac{1}{4} (2 + 4e^{-j\frac{\pi}{2}k}) \\ &= \frac{1}{2} (1 + e^{-j\frac{\pi}{2}k}) \end{aligned}$$

(5)

$$\begin{aligned} a_k &= \frac{1}{4} [1 + (1 + \frac{\sqrt{2}}{2})e^{-j\frac{\pi}{2}k} + (1 + \frac{\sqrt{2}}{2})e^{-j\frac{3\pi}{2}k}] \\ &= \frac{1}{4} [1 + (2 - \sqrt{2})\cos k\frac{\pi}{2}] \end{aligned}$$

4-2

(1) $N=8$

$$\begin{aligned} x[n] &= \sum_{k=0}^7 a_k e^{jk\frac{\pi}{4}n} \\ &= \sum_{k=0}^7 (\cos \frac{k}{4}\pi + \sin \frac{3}{4}k\pi) e^{jk\frac{\pi}{4}n} \end{aligned}$$

(3)

$$\begin{aligned} x[n] &= \sum_{k=-1}^2 a_k e^{jk\frac{\pi}{4}n} \\ &= e^{-j\frac{\pi}{4}n} + 1 + e^{j\frac{\pi}{4}n} \\ &= 1 + 2\cos k\frac{\pi}{4} \end{aligned}$$

(5)

$$\begin{aligned} X[z] &= \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{\pi}{2}n} \\ &= a_0 + a_4 e^{j2\pi n} + a_1 e^{j\frac{\pi}{2}n} + a_5 e^{j\frac{5\pi}{2}n} \\ &\quad + a_2 e^{j\pi n} + a_6 e^{j3\pi n} + a_3 e^{j\frac{3\pi}{2}n} \\ &\quad + a_7 e^{j\frac{7\pi}{2}n} \\ &= (a_0 + a_4) + (a_1 + a_5) e^{j\frac{\pi}{2}n} \\ &\quad + (a_2 + a_6) e^{j\pi n} + (a_3 + a_7) e^{j\frac{3\pi}{2}n} \end{aligned}$$

$$X[n] = \begin{cases} 0 & n \text{ 偶} \\ \frac{1}{4} & n \text{ 奇} \end{cases}$$

4-3

$$\begin{aligned} (1) \quad x[n] + x[n + \frac{N}{2}] &= 0 \\ \sum_{k=-\infty}^{\infty} a_k [(1 + (-1)^k)] e^{jk\frac{\pi}{2}n} &= 0 \end{aligned}$$

$$\begin{aligned} \therefore a_k [(1 + (-1)^k)] &= 0 \\ a_{2k} &= 0 \end{aligned}$$

(2)

$$\begin{aligned} x[n] + x[n + \frac{N}{4}] &= 0 \\ a_k (1 + e^{jk\frac{\pi}{4}}) &= 0 \end{aligned}$$

$$k \text{ 能被 } 4 \text{ 整除时 } e^{jk\frac{\pi}{4}} = 1 \quad \therefore a_k = 0 \quad a_k = 0$$

$$\therefore a_{4k} = 0 \quad k \text{ 任意整数}$$

4-4 (1)

$$\begin{aligned} x[n] &= \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{\pi}{4}n} \\ x^*[n] &= \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\frac{\pi}{4}n} = \sum_{k=-\infty}^{\infty} a_{-k}^* e^{jk\frac{\pi}{4}n} \end{aligned}$$

$$a_k = a_{-k}^* \quad b_k + jc_k = b_{-k} - jc_{-k}$$

$$b_k = b_{-k}$$

$$c_k = -c_{-k}$$

(2)

$$\text{奇: } C_{\frac{N}{2}} = -C_{-\frac{N}{2}}$$

$$\text{周期: } C_{\frac{N}{2}} = C_{\frac{N}{2}}$$

$$\therefore C_{\frac{N}{2}} = 0 \quad a_{\frac{N}{2}} \text{ 实数}$$

4-15

$$\begin{aligned}
 (1) \quad x[n]y[n] &= \sum_{l=-\infty}^{\infty} a_l e^{j\frac{2\pi}{N}nl} \sum_{m=-\infty}^{\infty} b_m e^{j\frac{2\pi}{N}mn} \\
 &= \sum_{l=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} a_l b_{k-l} e^{j\frac{2\pi}{N}nk} \quad (k=m+l) \\
 &= \sum_{k=-\infty}^{\infty} \left(\sum_{l=-\infty}^{\infty} a_l b_{k-l} \right) e^{j\frac{2\pi}{N}nk} \\
 c_k &= \sum_{l=-\infty}^{\infty} a_l b_{k-l}
 \end{aligned}$$

$$\begin{aligned}
 c_k &= \sum_{l=-\infty}^{\infty} a_l b_{k-l} = \sum_{n=-\infty}^{\infty} x[n] y[n] e^{-j\frac{2\pi}{N}nk} \cdot \frac{1}{N} \\
 k=0 \quad \int \sum_{n=-\infty}^{\infty} x[n] y[n] &= \sum_{l=-\infty}^{\infty} a_l b_{-l} \cdot N
 \end{aligned}$$

(2)

①

$$\cos \frac{6\pi}{N}n = \frac{1}{2} (e^{j\frac{6\pi}{N}n} + e^{-j\frac{6\pi}{N}n})$$

$$b_3 = b_{-3} = \frac{1}{2}$$

$$c_k = \sum_{l=0}^{N-1} a_l b_{k-l}$$

$$= \frac{1}{2} (a_{k-3} + a_{k+3})$$

②

$$b_k = \frac{1}{N}$$

$$c_k = \frac{1}{N} \sum_{l=-\infty}^{\infty} a_l$$

(3)

$$x[n] = \frac{1}{2} (e^{j\frac{\pi}{3}n} + e^{-j\frac{\pi}{3}n})$$

$$b_2 = b_{-2} = \frac{1}{2}$$

$$a_k = \sum_{n=-\infty}^{\infty} y[n] e^{-j\frac{\pi}{6}kn} \cdot \frac{1}{12}$$

$$\begin{aligned}
 &= \left(e^{-j\frac{\pi}{2}k} + e^{j\frac{\pi}{2}k} + e^{-j\frac{\pi}{3}k} + e^{j\frac{\pi}{3}k} + e^{-j\frac{\pi}{6}k} + e^{j\frac{\pi}{6}k} + 1 \right) \times \frac{1}{12} \\
 &= \frac{1}{6} (\cos \frac{\pi}{2}k + \cos \frac{\pi}{3}k + \cos \frac{\pi}{6}k + 1) \cdot \frac{1}{2}
 \end{aligned}$$

$$c_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}$$

$$= \sum_{k=-\infty}^{\infty} \frac{a_{k-2} + a_{k+2}}{2}$$

$$= \frac{1}{12} (\cos \frac{\pi}{6}k - \cos \frac{\pi}{3}k - 2\cos \frac{\pi}{2}k + 1)$$